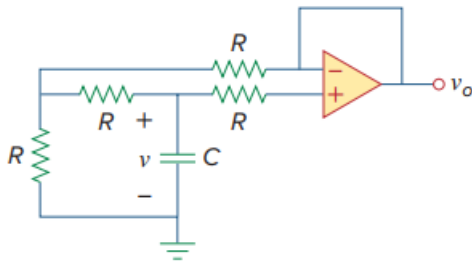


Problem

If $v(0) = 10 \text{ V}$, find $v_o(t)$ for $t > 0$ in the op amp circuit in Fig. 7.132. Let $R = 100 \text{ k}\Omega$ and $C = 20 \mu\text{F}$.

Fig. 7.132:



Step-by-step solution

Step 1 of 4

Refer to Figure 7.132 in the textbook.

For an ideal OPAMP, current through the OPAMP input is zero. So, Resistor R connected to non-inverting input can be neglected.

Draw the modified circuit diagram.

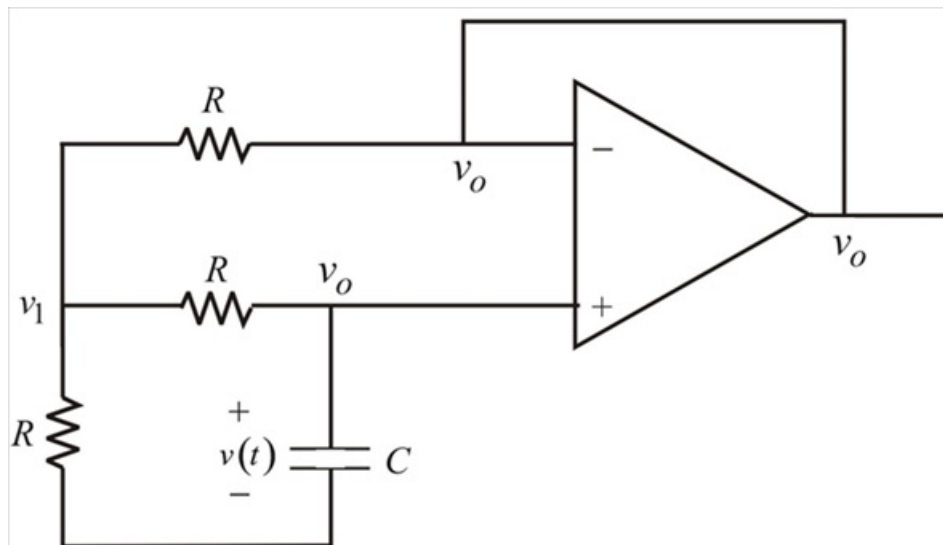


Figure 1

Step 2 of 4

Apply Kirchhoff's current law at node, v_1 .

$$\frac{v_1}{R} + \frac{v_1 - v_o}{R} + \frac{v_1 - v_o}{R} = 0$$

$$v_1 \left(\frac{1}{R} + \frac{1}{R} + \frac{1}{R} \right) = v_o \left(\frac{1}{R} + \frac{1}{R} \right)$$

$$v_1 \left(\frac{3}{R} \right) = v_o \left(\frac{2}{R} \right)$$

$$v_1 = \frac{2}{3} v_o$$

The capacitor voltage is same as the output voltage.

$$v_C = v_o$$

The current through the capacitor is,

$$\begin{aligned} i_C &= C \frac{dv_C}{dt} \\ &= C \frac{dv_o}{dt} \end{aligned}$$

Step 3 of 4

Apply Kirchhoff's current law at non-inverting terminal

$$i_C + \frac{v_o - v_1}{R} = 0$$

$$C \frac{dv_o}{dt} + \frac{v_o - v_1}{R} = 0$$

Substitute $\frac{2}{3} v_o$ for v_1 .

$$C \frac{dv_o}{dt} + \frac{v_o - \frac{2}{3} v_o}{R} = 0$$

$$C \frac{dv_o}{dt} + \frac{1}{3R} v_o = 0$$

$$\frac{dv_o}{dt} + \frac{1}{3RC} v_o = 0$$

Step 4 of 4

The output voltage from the differential equation is,

$$\begin{aligned} v_o(t) &= v_o(0) e^{-\frac{t}{3RC}} \\ &= v(0) e^{-\frac{t}{3RC}} \end{aligned}$$

Substitute 10 V for $v(0)$, 100 k Ω for R and 20 μ F for C .

$$\begin{aligned} v_o(t) &= 10 e^{-\frac{t}{3(100 \times 10^3)(20 \times 10^{-6})}} \\ &= 10 e^{-\frac{t}{6}} \text{ V} \end{aligned}$$

Thus, the expression for the output voltage is $\boxed{10 e^{-\frac{t}{6}} \text{ V}}$.